

Stable Landmark Identity and Unlabeled Adaptation: A ConvNeXt Framework for FoundUS 2026

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Abstract. Reliable ultrasound biometry requires accurate landmark localization, stable landmark identity, and robustness to acquisition variation. We present a ConvNeXt framework for the FoundUS 2026 challenge that combines stable landmark identity with adaptation from unlabeled data. A ConvNeXt Small encoder with separate heatmap heads predicts landmarks across nine measurement tasks. Ordered endpoint identities remain fixed during training and model aggregation, with a deterministic conversion applied at the output boundary. In a controlled adaptation study, residual adapters learn from labeled images whose appearance statistics are transferred from unlabeled images of the same task. Across five folds grouped by sequence, the combined proxy improved in every fold relative to a matched labeled donor control, reducing macro mean radial error (MRE) by 0.0367 pixels and measurement proxy mean absolute error (MAE) by 0.0153. For the final challenge system, an ultrasound foundation model supports distillation of ensemble predictions on the official unlabeled pool into a single deployable model. The final model achieved official validation MRE and MAE of 24.7986 and 25.9233 and hidden test values of 24.9356 and 26.8758. These results support stable landmark identity and complementary use of unlabeled data in a practical ultrasound biometry framework.

Keywords: Ultrasound biometry · Landmark localization · Stable landmark identity · Unlabeled adaptation · ConvNeXt · Ultrasound foundation model

1 Introduction

Ultrasound biometry supports assessment across fetal, intrapartum, and cardiac imaging by converting anatomical observations into interpretable measurements. Automated analysis may reduce manual effort and improve reproducibility, but ultrasound remains difficult because image formation is affected by speckle, acoustic shadows, operator technique, device settings, and incomplete depiction of anatomy. Learning systems have shown potential for standard plane recognition, fetal measurement, and cardiac functional assessment [2, 3, 11, 9].

Many biometric quantities are derived from multiple anatomical landmarks rather than a single region. Reliable landmark localization therefore depends on both spatial accuracy and stable landmark identity for each point. This distinction is important for ordered endpoints: two endpoint assignments may yield the same derived distance while representing different anatomical correspondence. Heatmap prediction is effective for anatomical landmark localization [10]. Reliable interpretation further depends on consistent correspondence between output channels and endpoint semantics.

A second challenge is the imbalance between limited manual annotation and abundant unlabeled ultrasound data. Semi-supervised learning can incorporate unannotated images through consistency learning and confidence guided pseudo-labeling [13, 12]. Ultrasound foundation models offer a complementary source of transferable representations learned from broader ultrasound collections [7]. For landmark prediction, these additional signals can complement supervised learning when anatomical correspondence remains stable and uncertainty in transferred information is appropriately controlled.

Within the FoundUS 2026 challenge setting [1], this work studies multitask ultrasound biometry through stable landmark identity and the use of unlabeled data. We examine controlled appearance adaptation and transferable ultrasound representations while maintaining anatomical correspondence across heterogeneous measurement tasks.

2 Method

2.1 Framework Overview

Let $\mathcal{T}$ denote the nine FoundUS task identifiers: A4C, AOP, FA, FUGC, HC, IVC, PLAX, PSAX, and fetal femur. For task $t \in \mathcal{T}$, the labeled collection contains an image x , landmark coordinates y , and a task identifier, while the unlabeled collection contains an image u and its task identifier. The framework uses a shared ConvNeXt heatmap predictor with a consistent landmark representation. Unlabeled data are incorporated through controlled appearance adaptation. For the final system, five fused predictors, one from each fold, generate distillation targets over the complete readable unlabeled pool, and inference uses a single fused student model.

Within the shared prediction pipeline, images are resized using aspect-ratio-preserving letterboxing, and landmark coordinates are transformed accordingly to the padded canvas. Each landmark is represented by a Gaussian heatmap. Batches contain samples from a single task and are routed through the corresponding prediction head and, when applicable, the task-specific adaptation module.

2.2 ConvNeXt Heatmap Predictor

The supervised predictor uses a ConvNeXt Small encoder [8] shared by all tasks. The final encoder feature is passed to a separate decoder head for each task. A

standard head applies convolutional refinement and progressive bilinear upsampling before producing one heatmap logit channel for every landmark. This design shares visual representation while retaining task specific output semantics.

For PSAX, the endpoint heatmap head is supplemented with auxiliary mid-point, tube, and direction predictions because its four endpoints define two measured line segments. These auxiliary targets provide structural supervision during training. Inference remains based on the endpoint heatmaps, preserving the same coordinate output interface as the other tasks.

For each heatmap channel, coordinate decoding uses a probability weighted average over its strongest responses rather than a single discrete maximum. The decoded coordinates are mapped through the inverse letterbox transform to the original image.

2.3 Stable Landmark Identity

Stable landmark identity is implemented through a deterministic representation convention. The apical four-chamber (A4C) and parasternal short-axis (PSAX) tasks contain ordered endpoints for which annotation order and numerical coordinate order do not always define a stable internal correspondence. We therefore establish a canonical channel order before optimization so that each heatmap channel retains a fixed endpoint definition. The A4C endpoints are organized into eight pairs and the PSAX endpoints into two pairs. For A4C, the first endpoint has the smaller vertical coordinate in odd numbered pairs and the smaller horizontal coordinate in even numbered pairs. For PSAX, the first endpoint has the larger horizontal coordinate. The same deterministic permutation is applied to normalized coordinates, original coordinates, and heatmap targets.

The canonical channel order is retained throughout training, checkpoint evaluation, and model aggregation. Fold predictions are aggregated in this shared channel space before a single deterministic conversion to the challenge output order. Additional canonicalization is applied only where the task definition requires an internal endpoint identity distinct from the output ordering; the remaining tasks retain their original channel definitions.

2.4 Task Specific Residual Adaptation

Controlled appearance adaptation starts from the trained ConvNeXt predictor. Its encoder and original heatmap heads remain fixed, and an independent residual adapter is assigned to each task. For task t , the adapter transforms the encoder feature z into

$$z'_t = z + sA_t(z), \quad (1)$$

where A_t denotes the residual adapter and s controls its contribution. Each adapter contains a pointwise channel reduction, normalization and nonlinear activation, a depthwise spatial convolution, and a pointwise expansion back to the encoder dimension. The expansion layer is initialized to zero, so the adapted predictor initially reproduces the supervised predictor.

Only the selected task adapter is optimized; the shared encoder and prediction heads remain fixed.

2.5 Task Conditioned Appearance Replay

In the appearance replay branch, manual annotations define the landmark targets, while unlabeled donors contribute task specific appearance statistics. A labeled image x is paired with an unlabeled donor u from the same task. Motivated by the mean and variance alignment used in adaptive instance normalization [5], we convert normalized tensors to RGB and compute per channel means and standard deviations inside the valid letterbox region. Donor statistics are grouped and averaged when the donor batch is larger than the content batch so that each loaded donor contributes to the replayed samples.

Let x^{rgb} and u^{rgb} denote the denormalized content and donor images. Let M be the valid content mask, and let μ_x, σ_x and μ_u, σ_u denote their masked channel moments. We interpolate the content moments toward the donor moments subject to bounded mean displacement and contrast ratio. The replayed RGB image is

$$\tilde{x}^{\text{rgb}} = M \odot \left(\sigma^* \frac{x^{\text{rgb}} - \mu_x}{\sigma_x} + \mu^* \right) + (1 - M) \odot x^{\text{rgb}}, \quad (2)$$

where μ^* and σ^* are the bounded target moments. The transformed image is clipped to the valid RGB range and normalized before prediction, while the padding region is copied unchanged.

Appearance replay changes photometric moments while preserving spatial content and coordinate geometry, so annotation y remains valid for both views.

2.6 Controlled Adaptation Objective

For controlled appearance adaptation, each update contains a clean labeled view and, when an unlabeled donor is available, its replayed counterpart. Both views use the same manual landmark annotation. Supervision is the mean squared error between sigmoid heatmaps and Gaussian targets. For PSAX, the midpoint, tube, and direction losses defined for the supervised predictor remain active for both views.

A reference consistency term regularizes the adapted clean prediction toward the corresponding prediction of the fixed supervised predictor. The objective is

$$\mathcal{L}_{\text{adapt}} = \mathcal{L}_{\text{clean}} + \lambda_r \mathcal{L}_{\text{replay}} + \lambda_a \mathcal{L}_{\text{ref}}, \quad (3)$$

where $\mathcal{L}_{\text{ref}}$ is the mean squared difference between the sigmoid heatmaps of the adapted model and the fixed supervised predictor. Gradients update only the routed residual adapter. For a task without a corresponding unlabeled pool, optimization follows the clean supervised branch.

2.7 Unlabeled Ensemble Distillation

For the final challenge system, we used a separate fused predictor built from the supervised ConvNeXt pathway and an ultrasound foundation model [7]. This final-system pathway is distinct from the residual-adapter model used in the controlled appearance adaptation study. The ConvNeXt and foundation-model task features are aligned at the heatmap resolution and combined by a residual fusion module, with the ConvNeXt prediction retained as a reference. The resulting fused predictor provides the basis for the final student. Its existing ConvNeXt pathway, foundation model decoder, fusion module, and base foundation model parameters remain fixed. Task-specific Low-Rank Adaptation (LoRA) modules [4] adapt later Transformer blocks of the foundation encoder, while a zero-initialized contextual correction head refines the reference heatmaps.

Distillation with teacher-generated targets on unlabeled examples provides the general learning setting [6]. In our final system, five fixed fold-specific fused predictors provide mean soft heatmap targets on the official unlabeled pool. Coordinate dispersion across teachers defines a continuous reliability weight applied to both heatmap KullbackLeibler (KL) divergence and decoded coordinate consistency.

3 Experiments

3.1 Experimental Setup

We evaluated nine FoundUS measurement tasks over five folds. The pooled held-out set contained 6,700 labeled images and 27,472 landmark points. Following the official data correction, 25 fetal femur images with inconsistent orientation were excluded. Explicit sequence identifiers were grouped where recoverable. The unlabeled pool contained 182,870 readable images after SHA-256 deduplication, with each image used once per fold in the donor schedule. Exact duplicates were removed, while related acquisitions could not be systematically identified from the available metadata. Fetal femur had no corresponding unlabeled pool.

The ConvNeXt input and heatmap resolutions were 518×518 and 64×64 . Residual adapters were optimized for eight epochs using AdamW with a learning rate of 10^{-4} , weight decay of 10^{-3} , bfloat16 precision, and seed 42. The labeled and donor batch sizes were 4 and 16. Appearance strength was sampled from $[0.25, 0.60]$, with mean displacement limited to 0.15 and the standard deviation ratio restricted to $[0.60, 1.60]$. The weak and strong fold-0 diagnostics used ranges of $[0.10, 0.30]$ and $[0.50, 0.80]$, respectively, and the reference-consistency weight was 0.25 by default and 0 in the corresponding diagnostic. Primary evaluation used epoch 8 and residual scale 1.0 for all folds, fixed before comparison.

The controlled study compared a fixed supervised reference with a matched labeled donor control and unlabeled donor adaptation. In both adaptation conditions, manual landmark targets came from the labeled content images, while same-task donor images supplied only appearance statistics. The two adaptation conditions used the same residual architecture, fold split, labeled schedule, and

optimization, with donor source as the comparison variable. A hard pseudo-label baseline used reference predictions on unlabeled images re-encoded as Gaussian heatmaps. Appearance strength, reference consistency, and residual scale were examined on fold 0 as diagnostics.

For the controlled study, we report task macro mean radial error (MRE), measurement proxy mean absolute error (MAE), and their arithmetic mean as a combined proxy. Official validation and hidden test results use the organizer’s Average MRE and Average MAE. For the final system, the complete readable unlabeled pool was used for distillation target generation, and inference used a single distilled model.

3.2 Results

Controlled appearance adaptation. Table 1 presents the pooled held-out results for the matched donor comparison. Unlabeled donor adaptation reduced macro MRE from 23.1183 to 23.0815 and measurement proxy MAE from 17.0117 to 16.9964, reducing the combined proxy from 20.0650 to 20.0390. Fold-wise changes were -0.0361 ± 0.0340 for MRE, -0.0154 ± 0.0099 for measurement MAE, and -0.0258 ± 0.0189 for the combined proxy (mean $\pm$ standard deviation); the combined proxy improved in all five folds.

A4C showed the largest improvement, whereas PSAX showed lower measurement MAE with slightly higher MRE. Changes in the remaining tasks were small, and fetal femur was unchanged because both conditions used the same supervised branch. Complete fold results are provided in the supplementary material. On official validation, the corresponding five-model ensembles achieved

Table 1. Pooled results over five folds for the matched donor comparison. Deltas denote unlabeled donor adaptation minus labeled donor control.

Task	MRE			Measurement MAE		
	Labeled	Unlabeled	Δ	Labeled	Unlabeled	Δ
A4C	26.6395	26.2897	-0.3498	14.9919	14.9403	-0.0516
AOP	4.4281	4.4288	+0.0006	2.4872	2.4879	+0.0007
FA	16.1339	16.1337	-0.0002	50.9719	50.9507	-0.0212
FUGC	6.4545	6.4522	-0.0023	4.0280	4.0280	0.0000
HC	19.6033	19.6044	+0.0011	19.5004	19.5117	+0.0114
IVC	49.4735	49.4705	-0.0030	13.1975	13.1794	-0.0181
PLAX	17.6175	17.6134	-0.0040	9.0588	9.0558	-0.0030
PSAX	47.2925	47.3194	+0.0268	22.1849	22.1291	-0.0557
Fetal femur	20.4217	20.4217	0.0000	16.6849	16.6849	0.0000
Macro	23.1183	23.0815	-0.0367	17.0117	16.9964	-0.0153

Average MRE values of 25.6454, 25.5334, and 25.4894 for the supervised reference, labeled donor control, and unlabeled donor adaptation, with Average MAE values of 26.7936, 26.2147, and 26.2592, respectively. Relative to the labeled donor control, unlabeled donor adaptation showed lower Average MRE and slightly higher Average MAE. The matched comparison over the held-out

folds therefore provides the primary controlled evidence for the contribution of unlabeled appearance.

Sensitivity and pseudo-label comparison. Under the same five fold protocol, the hard pseudo-label baseline achieved an MRE of 23.1593, measurement MAE of 16.9242, and combined proxy of 20.0417. Compared with unlabeled appearance adaptation, it showed higher MRE and lower measurement MAE, with a slightly higher combined proxy.

On fold 0, the default appearance range gave a lower combined proxy than the weak and strong variants, and removing reference consistency increased the proxy. The residual-scale response was non-monotonic; the primary five fold comparison retained the unit scale fixed before comparison. Complete sensitivity results are provided in the supplementary material.

The matched donor conditions used identical fold-specific schedules (12,552–12,568 steps) and 899,136 trainable adapter parameters. Recorded unlabeled donor runs required 21.0–25.1 minutes per fold on an RTX 5090; available wall-clock records are summarized in the supplementary material.

Landmark identity audit. Using fixed held-out predictions from the supervised reference, a pairwise correspondence audit covered 5,736 endpoint pairs. Allowing a within-pair swap only for analysis reduced MRE by 29.99 pixels for A4C horizontal pairs and 5.44 pixels for PSAX, compared with 0.01 pixels for A4C vertical pairs and 0.27 pixels for other paired tasks. Endpoint swapping left the derived measurement values and measurement MAE unchanged. These results support the distinction between landmark correspondence and measurement accuracy.

Final challenge system. Table 2 summarizes official validation performance of the supervised ConvNeXt ensemble, the teacher ensemble, and the final distilled model. The final model achieved Average MRE of 24.7986 and Average MAE of 25.9233, both lower than the supervised ConvNeXt ensemble.

Table 2. Official validation performance of the reference systems and final model.

System	Models at inference	Average MRE	Average MAE
Supervised ConvNeXt ensemble	5	25.6454	26.7936
Five-model fused teacher ensemble	5	24.0141	26.6722
Final fused student	1	24.7986	25.9233

The final model showed higher MRE and lower MAE than the teacher ensemble. On the hidden test set, the same final model achieved an Average MRE of 24.9356 and Average MAE of 26.8758.

4 Discussion

The framework addresses stable landmark identity and unlabeled appearance adaptation through distinct mechanisms. The identity convention maintains

endpoint semantics during training and aggregation, while appearance replay exposes residual adapters to task-related appearance variation. Under matched optimization, unlabeled donor adaptation produced a modest gain over the labeled donor control, with the combined proxy improving in all five folds. Clinical interpretation would require physical calibration, estimates of observer variability, and task-specific thresholds beyond the challenge metrics.

Effects varied by task: A4C contributed most of the gain, other tasks changed little, and PSAX showed lower measurement MAE but higher landmark MRE. PSAX auxiliary supervision was fixed across the controlled donor conditions, making donor source the varying factor in that comparison. Its individual contribution requires a dedicated ablation. The identity audit showed that endpoint correspondence can substantially alter MRE while leaving the derived measurement unchanged. Together, these observations indicate that localization accuracy, landmark correspondence, and derived measurement accuracy are related but distinct properties of biometric prediction. The audit therefore supports preserving stable landmark identity for ordered endpoints, while task-specific differences reflect each measurement problem.

The controlled study used one random seed and folds grouped by recoverable sequence identifiers. SHA-256 deduplication removed exact duplicates; available metadata did not permit systematic identification of all related acquisitions. Fetal femur had no corresponding unlabeled pool and retained the supervised branch. Use of the complete challenge unlabeled pool places these experiments within the transductive FoundUS setting. Independent cohorts, devices, and repeated training runs would provide stronger evidence beyond this setting.

For the final challenge system, supervised fusion combined ConvNeXt and ultrasound foundation representations, and subsequent unlabeled distillation compressed the five-model teacher ensemble into a single inference model. Aggregate validation MRE and MAE were lower than for the supervised ConvNeXt ensemble, while comparison with the teacher ensemble again showed different MRE and MAE behavior. Because several components change together in the final system, matched ablations would be required to isolate their individual contributions.

5 Conclusion

We presented a ConvNeXt framework for multitask ultrasound biometry with stable landmark identity and complementary use of unlabeled data. Controlled appearance adaptation yielded small, consistent gains under matched validation, and the identity audit showed that endpoint correspondence contains information beyond derived measurements. The final system combined foundation model fusion and unlabeled ensemble distillation, achieving lower official validation MRE and MAE than the supervised ConvNeXt ensemble. These results support consistent landmark representation and controlled use of unlabeled ultrasound data.

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